

1. Basic sound setup

About **\$450** with a borrowed calibrator; **\$700** with our own.

Use our computer and neural recorder. This is the simplest proposed setup for selected tones below 20 kHz. The microphone checks what reaches the ear.

Computer → **USB interface** → **powered speaker**

Microphone near ear → **interface** → **sound recording**

Parts to buy

Blue names are buying links. USD asking-price snapshot: October 9, 2026.

Job	Part / seller	Price
Play and record	Behringer UMC204HD – eBay / Sweetwater, new	97.98
Make the sound	One JBL 305P MkII powered speaker – Reverb, new/open box	122.39
Measure sound	Dayton EMM-6 microphone – eBay / Parts Express, new	66.93
Mic cable	Hosa HMIC-010 XLR cable – Sweetwater	17.99
Audio cables	Two Hosa HSS-005 TRS cables , \$14.95 each	29.90
Hold the mic	Gator RI-MICTP-FBM stand – Sweetwater	34.99
Parts	Borrow the sound-level calibrator	370.18
Optional purchase	REED R8090 calibrator – DigiKey / REED, new	219.00

Working totals: \$370.18 + \$80 shipping/tax reserve = **\$450.18**. With our own calibrator: \$589.18 + \$100 reserve = **\$689.18**. A recorder-specific cable/adaptor, if needed, is additional.

What could be misaligned?

Start delay. Buffering alone can add several milliseconds. One 256-sample buffer at 48 kHz lasts **5.3 ms**; two last 10.7 ms. Converter, speaker and air delays are additional. This is an example, not measured UMC latency. Air adds about **3 ms per metre** in either setup.

Trial variation. How much the delay changes between trials is **unmeasured**. Preload the sequence and use wired audio. The moment the program calls “play” is not the sound’s arrival time.

Clock drift. The interface-to-recorder mismatch is **unknown**. For scale, a hypothetical **100 ppm relative mismatch** means **6 ms per minute, or 0.36 seconds per hour**, if uncorrected. This is an illustration, not a prediction for this hardware.

How we align it

Generate a reference alongside each tone on the spare output and record it with the brain data. Measure its offset to sound arrival with the microphone. Recording the mic only on the USB interface still leaves two clocks to align. Audio output is not TTL; recorder inputs determine the adapter. A compatible neural-recorder mic input would give a more direct acoustic reference.

Timing sources: [PortAudio buffer calculation](#); [latency estimates](#); [NTi acoustic delay](#). Sound level still needs per-tone calibration at the ear.

2. Full SONIC sound range

About **\$11,000–\$12,000**, pending the TDT quote.

This buys a proposed 1.4–32 kHz sound and measurement chain, plus hardware-timed playback. It does not, by itself, establish lower drift against our neural recorder.

Parts and cost

Job	Part / buying status	Price
Timed sound I/O	NI USB-6361, 781442-01 – Newark, new	3,030.42
Speaker	TDT MF2-M kit – quote / order	1,500 est.
Amplifier	TDT SA1 – quote / order	2,000 est.
Amplifier power	TDT ZB1PS – quote / order	750 est.
Measurement mic	Earthworks M50 – eBay, new	1,299.00
Mic preamp	Sound Devices MM-1 – eBay, used / no returns	325.00
Level reference	REED R8090 – same calibrator; count once	219.00
Connections	Cables, power and fixtures – allowance	500.00
Subtotal	Includes unquoted TDT allowances	9,623.42

Add 15% for shipping, tax and uncertainty: **\$11,066.93**. A new \$945 MM-1 instead gives **\$11,779.93**. TDT figures are estimates, not vendor prices. Labor, any extra clock distribution and specialist calibration service are excluded. This is not a fixed upper limit.

How much drift should we expect?

There is no measured Rice drift value yet. NI lists 50 ppm timing accuracy, under its default typical-at-25 °C conditions. These calculations show clock-error scale before synchronization or correction:

Clock comparison	After 1 min	After 1 hour
NI at 50 ppm versus ideal time	3 ms	180 ms
Hypothetical NI and recorder at opposite 50 ppm errors	6 ms	360 ms

The second row assumes a recorder specification we have **not confirmed**. Neither row is a guaranteed bound or a prediction of our alignment. A shared start trigger alone does not prevent drift.

Prevent accumulating mismatch by sharing a compatible clock, or use repeated recorded references to map the timelines. USB-6361 accepts a 10 MHz reference on PFI; recorder compatibility is open. Shared clocking still leaves acoustic delay and timing-measurement error.

NI specifications, pp.1, 6, 14; NI synchronization. Drift in milliseconds = relative error in ppm × elapsed seconds / 1,000.

Which would I start with?

The basic setup, if tones below 20 kHz answer our question. Test its alignment first. A proposed goal is **1 ms or less residual error after correction**; this is not achieved or guaranteed. Choose the expensive chain for the full SONIC frequency range. Arduino/FPGA can control playback and markers; sound arrival and drift still need measurement.

SONIC v1, Methods 6.2, p.11 used USB-6361, SA1, MF1 and PTP/10 MHz synchronization. It reports no measured acoustic-to-neural jitter or long-session drift. Rice's sedated sheep remain an adaptation of SONIC's awake, alert sheep. No setup here has been built or tested.